STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh
Program: Ph.D. in Electrical, Electronics and Computer Engineering
Institution: University of Michigan-Dearborn

Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy Laboratory focuses on distributed mathematical optimization for energy management. While this foundation drove my thesis on microgrid resilience, I recognize that classical optimization faces critical limitations when scaling to the real-time demands of modern grids. To overcome these bottlenecks, I seek to deepen my expertise in scalable cyber-physical control systems. UM-Dearborn's Ph.D. program provides the exact framework needed to architect decentralized grids for high renewable penetrations.

Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering instilled in me a systems-level perspective, training me to deconstruct complex power networks into dynamic, tractable optimization problems. This mathematical foundation drives my independent research trajectory, allowing me to architect solutions that bridge abstract algorithms with real-world physical constraints.

In research submitted to *Sustainable Energy, Grids and Networks* (SEGAN), I developed a peer-to-peer trading framework that subdivides distribution networks into autonomous microgrids, integrating rigorous grid physics (AC-OPF) with distributed optimization (ADMM). I validated this architecture across both radial and ring-connected topologies, proving it reduced operational costs while strictly bounding voltage levels within the $\pm 5\%$ threshold and preventing local trading from destabilizing the public grid. However, this static framework lacked temporal foresight to handle sudden failures.

To enhance grid resilience and achieve dynamic fault tolerance, my graduation thesis layered Model Predictive Control (MPC) over this baseline. By coordinating emergency power exchanges during simulated outages, this temporal control empowered neighboring networks to assist a failing grid, reducing forced load shedding from 15 MWh to just 3 MWh while protecting 100% of critical loads.

Overcoming strict convergence challenges and collaborating across research groups forged my resilient analytical foundation. Equipped with this physics-based modeling capability, I am prepared to drive scalable, real-time grid operations within your doctoral ecosystem.

Research Interests

My primary research objective is to engineer autonomous power systems with high survivability against catastrophic failures. To achieve this, I focus on advancing grid resilience through dynamic network reconfiguration and rapid fault response. Classical optimization serves as the rigorous mathematical foundation for these self-healing mechanisms, guaranteeing strict operational safety even during severe outages.

Building upon this physics-based foundation, my future research expands into Multi-agent Reinforcement Learning (MARL). I aim to leverage these learning-based policies as computational accelerators to bypass strict mathematical assumptions in classical solvers—specifically, the exactness issues of SOCP relaxations and the limitations of thermal constraints. This AI-driven synergy enables instantaneous, decentralized dispatch without sacrificing physical safety.

Academic and Research Alignment

The University of Michigan-Dearborn provides the rigorous theoretical coursework and collaborative research environment required to pursue my synergistic vision. Specifically, my trajectory aligns directly with the laboratory of Prof. Van-Hai Bui, driven by a shared focus on autonomous multi-agent energy management.

Building upon the deterministic optimization foundation detailed previously, I aim to contribute to his learning-based frameworks, such as FairMarket-RL and XRL-LLM, by embedding my mathematical models as physics-informed safety filters.

Beyond Prof. Bui's laboratory, the expertise of Dr. Wencong Su and Dr. Junho Hong strongly complements my holistic research vision. Dr. Su's research on utilizing LLMs for Economic Dispatch would deepen my understanding of system-level AI integration and large-scale energy management. Conversely, Dr. Hong's focus on physics-informed anomaly detection would strengthen my foundation in component-level physical constraints and grid security. Fusing these perspectives expands my research horizon, providing the deep insights necessary to engineer autonomous grid architectures that dynamically scale through real-time volatility while preserving strict physical safety guarantees.

Career Objectives

Following the completion of my Ph.D, I intend to pursue a career as an R&D scientist dedicated to the advancement of global power infrastructures. I am deeply committed to applying these theoretical breakthroughs to solve high-impact challenges in developing regions, specifically driving the technical modernization and resilience of Vietnam's national grid.